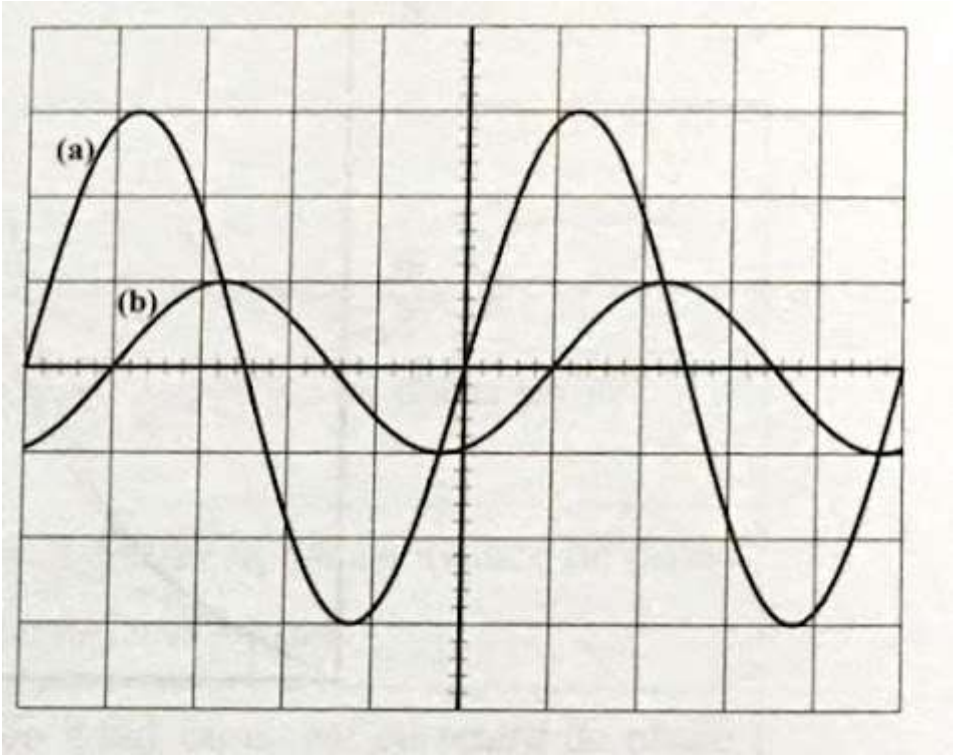
An oscilloscope displays two voltages from the same electrical circuit as shown in the figure below. For both channels: Sv = 5 V/div and Sh = 2 ms/div.

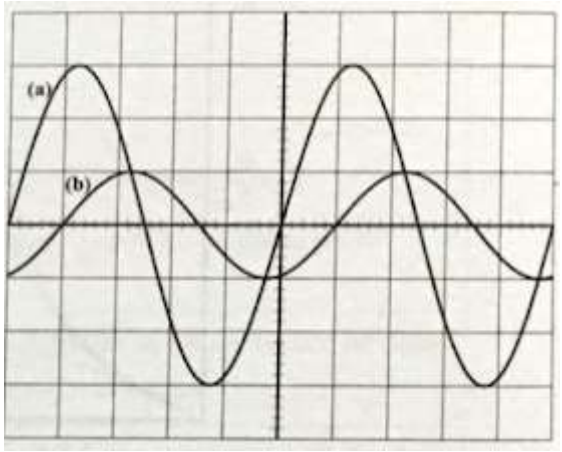



2. How many divisions is the amplitude (Ymax) of Oscillogram (a)? 1 point

*Mark only one oval.*

- 1 div
- 2 div
- 3 div
- 4 div

3. So what is the maximum voltage for voltage (a)? 1 point




*Mark only one oval.*

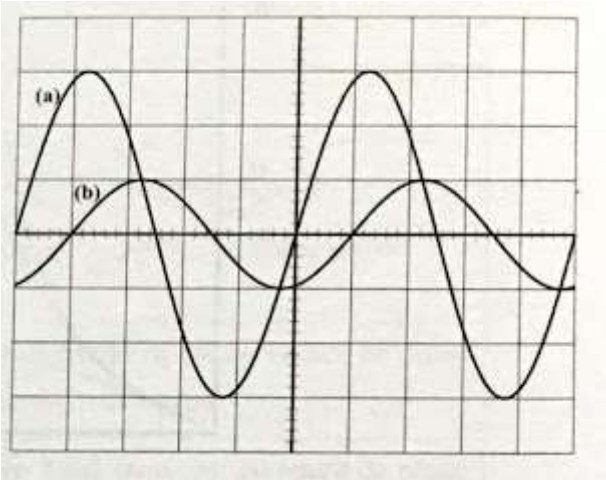
- 5 V
- 10 V
- 15 V
- 20 V

4. What is the effective value of visualized voltage (a)? 1 point

*Mark only one oval.*

- 5.6 V
- 10.6 V
- 15.6 V
- A

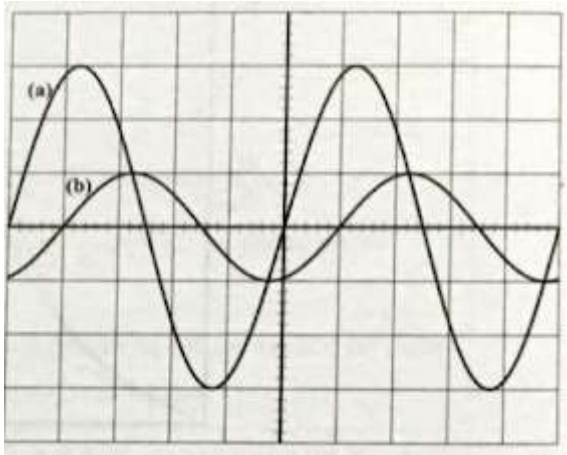
5. Calculate the effective value of visualized voltage (b). 1 point




*Mark only one oval.*

- 3.54 V
- 5.54 V
- 10.54 V
- 12.54 V

6. Calculate the period of the generator. 1 point




*Mark only one oval.*

- 5 ms
- 6 ms
- 8 ms
- 10 ms
- 12 ms

7. Deduce the generator's frequency. 1 point

*Mark only one oval.*

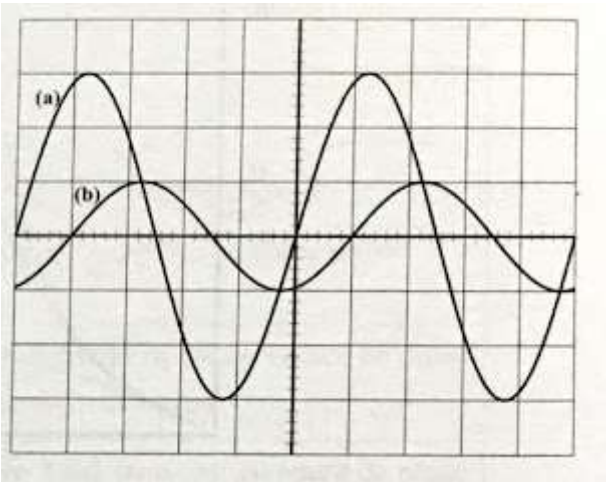
- 1 Hz
- 2 Hz
- 10 Hz
- 50 Hz
- 60 Hz
- 100 Hz
- 1000 Hz

8. So what is the pulsation ω? 1 point

*Mark only one oval.*

- ω = π rad/s
- ω = 2π rad/s
- ω = 5π rad/s
- ω = 10π rad/s
- ω = 100π rad/s
- ω = 150π rad/s
- ω = 200π rad/s

9. Which of the oscillograms between (a) or (b) is in advance (leads) to the other? 1 point



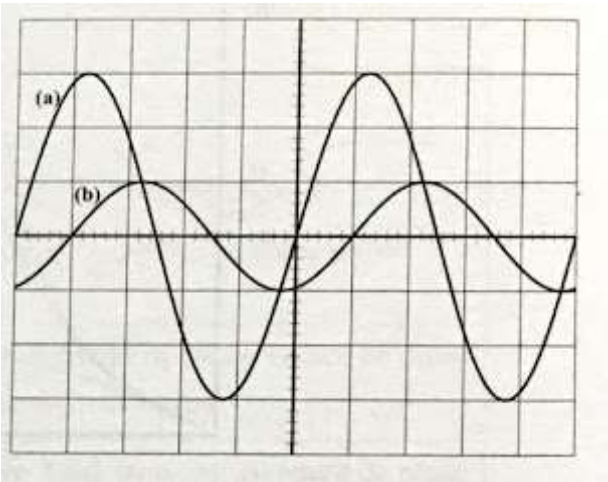

*Mark only one oval.*

- (a) leads (b)
- (b) leads (a)
- (a) lags behind (b)
- None of them: (a) and (b) are in phase

Hint for finding: phase difference

$$\varphi = \frac{2\pi\tau}{T}$$

10. Using the hint above, find the phase-shift or phase difference between (a) and (b). 1 point




*Mark only one oval.*

- φ = 0 (rad)
- φ = π/5 (rad)
- φ = π/4 (rad)
- φ = π/2 (rad)
- φ = 2π/5 (rad)
- φ = π (rad)
- φ = 2π (rad)

11. Knowing that the expression of the voltage represented by the oscillogram (a) is: 1 point

$$u_a = U_{ma} \cos(\omega t)$$

*Mark only one oval.*

$u_b = 2\cos(2\pi t)$

Option 1

$u_b = 5\cos(2\pi t)$

Option 2

$u_b = 5\cos(200\pi t)$

Option 3

$u_b = 5\cos(200\pi t + \frac{2\pi}{5})$

Option 4

$u_b = 5\cos(200\pi t - \frac{2\pi}{5})$